AI Hackathon 2026

30 more project ideas for a Jac + Jaseci build (graphs, walkers, by llm)

How to read this

These ideas are a second batch, chosen to avoid the most common hackathon picks (government-services navigator, study planner, farm advisor, disaster alert, shop chatbot). Every idea is shaped around the three things Jac does best: data as a graph, walkers that move through it, and by llm for reasoning and writing.

Build effort and novelty are my own estimates, not measured. Market demand has not been verified.

Rules to keep in mind (from the event documents)

- At least 80% of the code must be Jac.
- All coding happens during hacking hours; brainstorming beforehand is fine and repos are checked.
- Demo video and submission are due strictly by 5:30 PM; late entries cannot win.
- If you use mock or synthetic data, say so clearly. Misrepresenting a project means disqualification.
- Scoring: technical execution, use of Jac and Jaseci, creativity, depth of agentic behaviour, impact and novelty, and the demo video.

What a strong idea needs

- A real graph, where relationships matter more than rows.
- A walker that makes visible decisions as it moves, ideally with a retry or self-check loop.
- A demo that changes on screen within 30 seconds.
- Scope that fits roughly 6.5 hours of real coding by three people.

Quick comparison

#	Idea	One-line pitch	Effort	Novelty
1	**TransformerTwin**	Predictive maintenance agent for distribution transformers	Med	High
2	**SolarSizer**	Rooftop solar and net-metering planner for households	Low	Med
3	**PowerQualityTracer**	Finds the source of voltage sags and harmonics inside a factory	Med	High
4	**NetlistDoctor**	Circuit and PCB netlist reviewer	Med	High
5	**MigrationMapper**	Database schema change impact analyser	Low	Med
6	**APIDriftWatch**	Detects breaking API changes and who they hurt	Low	Med
7	**OnboardBuddy**	Turns any repo into a personalised learning path	Med	Med
8	**TestGapHunter**	Finds untested critical code paths and writes the tests	High	Med
9	**CloudCostCrawler**	Finds wasted cloud resources and explains why	Low	Med
10	**InvoiceChaser**	Late-payment recovery agent for freelancers and SMEs	Low	Med
11	**SupplierRiskMap**	Hidden single-point-of-failure finder for SME supply chains	Med	High
12	**SkillMatch**	Explainable CV-to-role matching	Low	Med
13	**MenuMath**	Menu engineering and ingredient-price-shock agent for restaurants	Low	Med
14	**FreelanceScope**	Scope-creep detector for freelancers	Low	Med
15	**MedInteract**	Household medicine interaction and schedule checker	Med	Med
16	**AllergenGuard**	Allergy-safe substitution for canteens and restaurants	Low	Med
17	**WorkSafeTrace**	Incident-to-root-cause agent for factories and construction sites	Low	High
18	**BloodLink**	Blood donor and hospital matching	Med	Med
19	**LabFault**	Fault-tree tutor for electronics lab circuits	Low	Med

#	Idea	One-line pitch	Effort	Novelty
20	**ConceptBridge**	Cross-module concept connector for university courses	Low	Med
21	**GradeAssist**	Rubric-based marking assistant for lecturers	Med	Med
22	**CampusFlow**	Timetable and room clash resolver	Med	Med
23	**PolicyDiff**	Plain-language impact tracker for new regulations	Low	Med
24	**DonationRouter**	Matches donated goods to verified needs	Med	Med
25	**VillageWater**	Rural water scheme fault and maintenance tracker	Low	High
26	**CharityVerify**	Claim checker for fundraising appeals	Med	Med
27	**RecipeRescue**	Cooks with whatever is left in your fridge	Low	Low
28	**StoryWeaver**	Children's story generator with consistent characters	Med	Med
29	**DebateCoach**	Argument-graph sparring partner	Med	Med
30	**BalanceBot**	Game balance tester using simulated players	Med	High

A. Electrical & Hardware

1. TransformerTwin

Predictive maintenance agent for distribution transformers

Gap: Utilities lose transformers to overload and nobody sees the pattern until it fails.

Graph: Substation → feeders → transformers → customers, with load history on each node.

Agents and walkers: LoadWalker traces how an overload propagates downstream; DiagnosisAgent (by llm) reads sensor trends and explains the likely failure mode; PlannerAgent schedules inspections by risk.

Demo: Replay a simulated heatwave day; transformers turn amber then red, and the agent ranks which to inspect first.

Watch out: Use synthetic load profiles and say so.

Effort / novelty: Med / High

2. SolarSizer

Rooftop solar and net-metering planner for households

Gap: Households cannot tell what system size, battery and payback actually fit them.

Graph: Appliances → daily load profile → roof area → panel/inverter options → tariff tiers.

Agents and walkers: LoadWalker builds the household profile from a typed appliance list; SizingAgent compares options; ROIAgent explains payback in plain Sinhala/English.

Demo: Type 'AC, fridge, 4 fans, water pump' and get 3 sizing options with a graph showing which appliances drive cost.

Watch out: Tariff numbers must be labelled as assumptions.

Effort / novelty: Low / Med

3. PowerQualityTracer

Finds the source of voltage sags and harmonics inside a factory

Gap: Plants see equipment trip but cannot tell which machine caused the disturbance.

Graph: Incomer → panels → motors/drives/loads, with event timestamps per node.

Agents and walkers: TraceWalker walks upstream from the affected load matching event times; HypothesisAgent proposes causes (motor start, VFD harmonics, capacitor switching).

Demo: Inject a sag at one drive; the walker lights the path back to the culprit.

Watch out: Synthetic waveforms are fine; keep the maths simple.

Effort / novelty: Med / High

4. NetlistDoctor

Circuit and PCB netlist reviewer

Gap: Students and small hardware startups ship designs with floating pins, missing decoupling and wrong pull-ups.

Graph: Components → pins → nets, parsed from a SPICE or KiCad netlist.

Agents and walkers: ParserWalker builds the graph; ReviewWalker applies checks along nets; FixAgent (by llm) proposes a corrected connection with reasoning.

Demo: Upload a flawed netlist; offending nets flash red with a suggested fix.

Watch out: Restrict to one netlist format.

Effort / novelty: Med / High

B. Developer Tools

5. MigrationMapper

Database schema change impact analyser

Gap: Changing a column breaks queries, reports and services nobody remembered.

Graph: Tables → columns → queries/services that read them.

Agents and walkers: ImpactWalker follows every reader of a changed column; RiskAgent scores each; MigrationAgent drafts a safe step-by-step migration.

Demo: Rename a column in a sample schema and watch affected services light up.

Watch out: Works on a sample schema; real data not needed.

Effort / novelty: Low / Med

6. APIDriftWatch

Detects breaking API changes and who they hurt

Gap: Teams publish an API change and find out which consumers broke from angry messages.

Graph: Endpoints → versions → consumer apps → fields used.

Agents and walkers: DiffWalker compares two OpenAPI specs; ConsumerWalker finds affected consumers; NoticeAgent writes a migration notice per consumer.

Demo: Paste two spec versions, get a red map of broken consumers and tailored emails.

Watch out: OpenAPI parsing is well-defined, so build risk is low.

Effort / novelty: Low / Med

7. OnboardBuddy

Turns any repo into a personalised learning path

Gap: New developers spend weeks learning which files matter.

Graph: Repo files/functions → call and import edges → concepts.

Agents and walkers: GuideWalker picks the shortest path to the part you must change; ExplainAgent explains each stop at your level.

Demo: Point it at the Jac repo; ask 'how do walkers run?' and it walks you through files in order.

Watch out: Pre-clone repos before the demo; avoid doing this during hacking hours without checking rules.

Effort / novelty: Med / Med

8. TestGapHunter

Finds untested critical code paths and writes the tests

Gap: Coverage numbers hide that the riskiest functions are never exercised.

Graph: Functions → call edges → test coverage markers → change frequency.

Agents and walkers: RiskWalker ranks critical untested nodes; TestAgent generates tests; ValidatorAgent runs them and retries on failure.

Demo: A coverage heatmap graph turns green as the agent adds tests.

Watch out: The run-and-retry loop is a strong agentic-depth demo.

Effort / novelty: High / Med

9. CloudCostCrawler

Finds wasted cloud resources and explains why

Gap: Startups pay for forgotten instances, orphaned volumes and idle databases.

Graph: Resources → owners → dependencies → usage metrics.

Agents and walkers: WasteWalker finds nodes with no inbound dependencies and low usage; SafetyAgent checks what would break if removed; SavingsAgent reports monthly savings.

Demo: Sample cloud account shows dead nodes greyed out with a 'safe to delete' verdict.
Watch out: Use a mock inventory and label it clearly.
Effort / novelty: Low / Med

C. Business & SME

10. InvoiceChaser

Late-payment recovery agent for freelancers and SMEs

Gap: Small businesses lose cash because chasing payments is awkward and manual.

Graph: Customers → invoices → payment history → relationship notes.

Agents and walkers: RiskWalker predicts who will pay late; ToneAgent drafts a reminder that fits the relationship; EscalationAgent decides when to step up.

Demo: Three late invoices get three very different messages.

Watch out: Messages should be drafts a human sends.

Effort / novelty: Low / Med

11. SupplierRiskMap

Hidden single-point-of-failure finder for SME supply chains

Gap: A shop or small factory does not realise two suppliers depend on the same port or vendor.

Graph: Products → ingredients/parts → suppliers → shipping routes.

Agents and walkers: DependencyWalker finds shared upstream nodes; ScenarioAgent simulates a disruption; AlternativeAgent proposes backups.

Demo: Trigger 'port delay' and watch affected products turn red.

Watch out: Synthetic supply data is acceptable.

Effort / novelty: Med / High

12. SkillMatch

Explainable CV-to-role matching

Gap: Hiring tools return a score but cannot explain it, and small firms cannot read 200 CVs.

Graph: Skills → related skills → roles → evidence in CV.

Agents and walkers: ExtractorAgent pulls skills with evidence; MatchWalker traverses related-skill edges to credit near-matches; ExplainAgent justifies every score.

Demo: Upload a CV; the graph shows why 'ROS' counts partly towards an embedded role.

Watch out: Flag fairness limits; never auto-reject.

Effort / novelty: Low / Med

13. MenuMath

Menu engineering and ingredient-price-shock agent for restaurants

Gap: When rice or coconut prices spike, owners do not know which dishes still make money.

Graph: Ingredients → dishes → cost → margin → popularity.

Agents and walkers: PriceWalker propagates a price change through every dish; MenuAgent suggests substitutions, portion changes or price moves.

Demo: Raise coconut price 40% and see which dishes lose margin.

Watch out: Prices are inputs the user types, so no scraping needed.

Effort / novelty: Low / Med

14. FreelanceScope

Scope-creep detector for freelancers

Gap: Clients keep adding work that was never in the agreement.

Graph: Contract deliverables → client requests → hours spent.

Agents and walkers: ScopeWalker checks each new request against the agreed deliverables; QuoteAgent drafts a polite change-order with a price.

Demo: Paste a chat with a client; out-of-scope requests are flagged with a drafted reply.
Watch out: Related to contract-risk ideas, so differentiate on the live chat angle.
Effort / novelty: Low / Med

D. Health & Safety

15. MedInteract

Household medicine interaction and schedule checker

Gap: Elderly people on several prescriptions get conflicting advice from different clinics.

Graph: Medicines → active ingredients → interactions → timing rules.

Agents and walkers: InteractionWalker finds conflicting pairs; ScheduleAgent builds a daily timetable; ExplainAgent writes simple warnings.

Demo: Enter five medicines and see a conflict edge highlighted.

Watch out: Strong disclaimer needed; it informs, never prescribes. Use a small verified interaction set.

Effort / novelty: Med / Med

16. AllergenGuard

Allergy-safe substitution for canteens and restaurants

Gap: Kitchens struggle to tell which dishes are safe for a given allergy and what to swap.

Graph: Dishes → ingredients → allergens → substitutes.

Agents and walkers: AllergenWalker traces hidden allergens through sub-recipes; SubstituteAgent proposes safe swaps that keep the taste.

Demo: Select 'no nuts' and the whole menu recolours.

Watch out: Use a small curated recipe set.

Effort / novelty: Low / Med

17. WorkSafeTrace

Incident-to-root-cause agent for factories and construction sites

Gap: Safety reports are filed but nobody links near-misses across sites.

Graph: Incidents → equipment → locations → shifts → causes.

Agents and walkers: PatternWalker links similar incidents; RootCauseAgent proposes a cause; ActionAgent drafts corrective actions.

Demo: Five scattered near-misses reveal one faulty guard.

Watch out: Synthetic reports only.

Effort / novelty: Low / High

18. BloodLink

Blood donor and hospital matching

Gap: Urgent blood requests spread over social media with no verification or compatibility check.

Graph: Donors → blood types → compatibility → hospitals → last-donation dates.

Agents and walkers: MatchWalker finds eligible nearby donors; OutreachAgent drafts personalised requests in Sinhala/Tamil; FollowUpAgent tracks replies.

Demo: Request O-negative at a hospital and eligible donors appear on the graph.

Watch out: Use fictional donors; do not store real personal data.

Effort / novelty: Med / Med

E. Education & Campus

19. LabFault

Fault-tree tutor for electronics lab circuits

Gap: Students cannot debug a breadboard circuit that does not work.

Graph: Symptoms → possible faults → measurements to take → conclusions.

Agents and walkers: DiagnoseWalker asks the student to measure a node and moves to the next likeliest fault; HintAgent explains why.

Demo: 'LED not lighting' leads to a guided path ending in a reversed diode.

Watch out: Good fit for the venue; keep to a few circuits.

Effort / novelty: Low / Med

20. ConceptBridge

Cross-module concept connector for university courses

Gap: Students do not see that Fourier in signals, control and communications is the same idea.

Graph: Concepts → modules → prerequisites → worked examples.

Agents and walkers: BridgeWalker finds where a concept reappears; ExplainAgent builds an analogy between modules.

Demo: Click 'convolution' and see it linked across four modules.

Watch out: Education space is crowded, so lead with the cross-module angle.

Effort / novelty: Low / Med

21. GradeAssist

Rubric-based marking assistant for lecturers

Gap: Marking engineering assignments is slow and inconsistent.

Graph: Rubric criteria → student answer sections → evidence.

Agents and walkers: EvidenceWalker maps each rubric item to a passage; GraderAgent scores and justifies; ConsistencyAgent flags outliers.

Demo: Mark three sample answers with explanations.

Watch out: Lecturer approves final marks; keep humans in the loop.

Effort / novelty: Med / Med

22. CampusFlow

Timetable and room clash resolver

Gap: Universities juggle rooms, lecturers and exams by hand.

Graph: Courses → lecturers → rooms → time slots → constraints.

Agents and walkers: ConflictWalker finds clashes; NegotiatorAgent proposes the smallest change set; ExplainAgent tells affected people.

Demo: Drag one lecture and watch the system repair the clash.

Watch out: Classic constraint problem; keep to a small dataset.

Effort / novelty: Med / Med

F. Civic & Social

23. PolicyDiff

Plain-language impact tracker for new regulations

Gap: People hear a new rule exists but not who it affects or what changes.

Graph: Regulation clauses → affected groups → previous version → obligations.

Agents and walkers: DiffWalker compares versions; ImpactAgent explains the effect per group (e.g. a small shop owner).

Demo: Paste two versions of a rule and see who is affected.

Watch out: Use public documents; present as summary, not legal advice.

Effort / novelty: Low / Med

24. DonationRouter

Matches donated goods to verified needs

Gap: After disasters, donations pile up in the wrong places.

Graph: Donors → items → shelters/needs → routes → stock.

Agents and walkers: NeedWalker finds the biggest unmet gaps; RouteAgent assigns pickups; UpdateAgent notifies donors.

Demo: Add a donation of blankets and see it routed to the shelter that needs them.

Watch out: Synthetic shelter data, labelled.

Effort / novelty: Med / Med

25. VillageWater

Rural water scheme fault and maintenance tracker

Gap: Community water schemes break down and nobody knows which part failed.

Graph: Source → pump → tank → pipelines → households.

Agents and walkers: FaultWalker traces which households lose water; MaintenanceAgent schedules repairs and drafts notices.

Demo: Break a pump and see downstream households go dark.

Watch out: Closely related to the feeder idea; good if the team likes network topology.

Effort / novelty: Low / High

26. CharityVerify

Claim checker for fundraising appeals

Gap: Donors cannot tell genuine appeals from scams.

Graph: Charity → claims → documents → registrations → past campaigns.

Agents and walkers: VerifyWalker cross-checks each claim; SkepticAgent flags inconsistencies; ReportAgent writes a trust summary.

Demo: Paste an appeal and get a claim-by-claim check.

Watch out: Use fictional organisations; avoid accusing real ones.

Effort / novelty: Med / Med

G. Consumer & Creative

27. RecipeRescue

Cooks with whatever is left in your fridge

Gap: Food gets wasted because people cannot see what to make from random leftovers.

Graph: Ingredients → recipes → substitutions → expiry dates.

Agents and walkers: PantryWalker finds recipes using soon-to-expire items; SubstituteAgent fills gaps; PlannerAgent sequences meals.

Demo: Enter five leftovers, get a three-day plan.

Watch out: Fun, visual and low risk.

Effort / novelty: Low / Low

28. StoryWeaver

Children's story generator with consistent characters

Gap: Generated stories forget characters and contradict themselves.

Graph: Characters → traits → relationships → plot events.

Agents and walkers: ContinuityWalker checks every new paragraph against the story graph; WriterAgent drafts; CriticAgent rejects contradictions and retries.

Demo: Story in Sinhala where a character's change in mood carries across chapters.

Watch out: Test Sinhala quality tonight.

Effort / novelty: Med / Med

29. DebateCoach

Argument-graph sparring partner

Gap: Students cannot see where their argument is weak.

Graph: Claims → evidence → rebuttals → assumptions.

Agents and walkers: ArgumentWalker finds unsupported premises; OpponentAgent attacks the weakest node; CoachAgent suggests repairs.

Demo: Paste an essay, watch weak nodes turn red, then debate it.

Watch out: Strong demo of agents interacting.

Effort / novelty: Med / Med

30. BalanceBot

Game balance tester using simulated players

Gap: Indie game developers cannot afford playtesters to find overpowered items.

Graph: Items → abilities → counters → win-rate edges.

Agents and walkers: PlayerAgents play thousands of simulated matches; BalanceWalker finds dominant nodes; DesignAgent proposes nerfs.

Demo: A rock-paper-scissors-style game gets rebalanced live.

Watch out: Pure simulation, no data issues; keep the game tiny.

Effort / novelty: Med / High